



Modelling Financial Markets

Chapter 17: Bubbles and Crashes



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Acronyms used in this material

- ▣ ADF: Augmented Dickey–Fuller (test)
- ▣ AI: Artificial Intelligence
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-FI: BET Financial Investment companies index
- ▣ BSADF: Backward Sup Augmented Dickey–Fuller (statistic)
- ▣ BVB: Bursa de Valori Bucureşti (the Bucharest Stock Exchange)
- ▣ CI: Confidence Interval
- ▣ ES: Expected Shortfall
- ▣ GSADF: Generalised Sup Augmented Dickey–Fuller (test)
- ▣ JLS: Johansen–Ledoit–Sornette (model)
- ▣ LPPL: Log-Periodic Power Law
- ▣ LPPLS: Log-Periodic Power Law Singularity (model)
- ▣ NVIDIA: NVIDIA Corporation (company name, US chip designer)
- ▣ OLS: Ordinary Least Squares
- ▣ PSY: Phillips–Shi–Yu (bubble test and date-stamping procedure)
- ▣ PWY: Phillips–Wu–Yu (bubble test and date-stamping procedure)
- ▣ SADF: Sup Augmented Dickey–Fuller (test)
- ▣ SE: Standard Error
- ▣ SRISK: Systemic RISK (capital shortfall measure)
- ▣ US: United States

Outline

Motivation

What Is a Bubble?

Run-Ups and Crashes in the Data

Testing for Explosive Behaviour

The Log-Periodic Power Law (LPPLS)

Regime-Switching Models

Crypto and AI Bubbles

AI for Scientific Discovery

Summary

References

Why Study Bubbles?

The question of this chapter

- ▣ Prices sometimes rise far faster than any plausible change in fundamentals, then lose most of their value
- ▣ Can we **detect** such episodes with statistics, and can we do it **in real time**?
- ▣ Can we say **when** they end?

What we do in this chapter

- ▣ Define a bubble: price minus the present value of future cash flows
- ▣ Test for explosive prices and date-stamp the episodes (PSY)
- ▣ Fit the LPPLS bubble model and check its crash-time forecasts honestly
- ▣ Describe booms and busts with regime-switching models; apply all tools to crypto and AI stocks

Tulip Mania, 1637



Hendrik Gerritsz Pot, *Flora's Wagon of Fools* (c. 1637), a satire of the tulip trade

Image: Hendrik Gerritsz Pot (c. 1637); public domain; [Wikimedia Commons](#)

- Dutch Republic, 1636–1637: contracts on rare tulip bulbs changed hands at very high prices, then the market stopped in February 1637
- The textbook “first bubble”; Garber argues that part of the price path is consistent with fundamentals of a new luxury good [Garber \(1990\)](#)
- Lesson 1: calling something a bubble requires a model of the fundamental value

The South Sea Bubble, 1720



William Hogarth, *The South Sea Scheme* (1721)

Image: William Hogarth (1721); CC0; Wikimedia Commons

- 1720: shares of the South Sea Company in London and of John Law's Mississippi Company in Paris rose several-fold within months and then collapsed [Garber \(1990\)](#)
- Common features: a new financial innovation, easy credit, a story about future profits, heavy trading [Kindleberger & Aliber \(2005\)](#)
- Lesson 2: bubbles come with credit and with high turnover [Scheinkman & Xiong \(2003\)](#)

The Crash of 1929



Crowd on Wall Street outside the New York Stock
Exchange, 29 October 1929

Photo: US government (1929); public domain; Wikimedia
Commons

- The 1920s boom in US stocks, partly financed with borrowed money, ended with the October 1929 crash and the Great Depression [Kindleberger & Aliber \(2005\)](#)
- Kindleberger's sequence (after Minsky): displacement, boom, euphoria, distress, panic [Kindleberger & Aliber \(2005\)](#)
- Lesson 3: we need tests that work on a single time series, in real time, before the panic

Data Used in This Chapter

Series	Frequency	Period	Use
S&P Composite, real price and dividend	monthly	1871–2024	price-dividend ratio
49 US industry portfolios	monthly	1926–2026	run-ups and crashes
Nasdaq 100, S&P 500	daily, monthly	1990–2026	dot-com, AI
BET, BET-FI	daily, monthly	1997–2026	Bucharest Stock Exchange
Bitcoin, Ethereum, Dogecoin	daily, weekly	2014–2026	crypto booms
NVIDIA, Cisco, Tesla, GameStop, Strategy	daily, monthly	1990–2026	single stocks

- Closing values up to 18 September 2026; stocks adjusted for dividends and splits; weekly and monthly series use the last price of the period
- Sources: shillerdata.com for the S&P Composite; [Kenneth French Data Library](https://www.kennethfrench.com) for industries

Plan of the Chapter

- ▣ **Theory:** fundamental value, rational bubbles, why they must explode, behavioural bubbles
- ▣ **Episodes:** nine run-ups and crashes in our data; what sharp run-ups predict (“Bubbles for Fama”)
- ▣ **Explosiveness tests:** right-tailed ADF, SADF, GSADF, BSADF date-stamping, critical values
- ▣ **LPPLS:** the model, its calibration and an honest check of its crash-time forecasts
- ▣ **Regimes:** Markov-switching models; **crypto and AI:** are we in a bubble?

Fundamental Value

- No-arbitrage with required return r : $P_t = \frac{\mathbb{E}_t[P_{t+1} + D_{t+1}]}{1 + r}$, with D the dividend
 - ▶ $\mathbb{E}_t$: expectation given the information available at t
- Solving forward gives the **fundamental value**: $F_t = \sum_{i \geq 1} \frac{\mathbb{E}_t[D_{t+i}]}{(1 + r)^i}$
 - ▶ Gordon case, dividends growing at $g < r$: $F_t = D_{t+1}/(r - g)$
- But $P_t = F_t + B_t$ also solves the equation whenever $\mathbb{E}_t[B_{t+1}] = (1 + r) B_t$
- B_t is the **rational bubble**: investors pay for it because they expect to resell it at a higher price [Blanchard & Watson \(1982\)](#)

Why a Rational Bubble Must Explode

- From $\mathbb{E}_t[B_{t+1}] = (1+r)B_t$: $\mathbb{E}_t[B_{t+h}] = (1+r)^h B_t$, an **explosive** path
 - ▶ The bubble grows faster than dividends ($r > g$), so its share of the price rises over time
- A negative bubble would eventually push the expected price below zero: rational bubbles are positive [Diba & Grossman \(1988\)](#)
- A bubble cannot start in a rational market: if it is zero today it must stay zero, so a bubble must exist from the first trade [Diba & Grossman \(1988\)](#)
- **Testable implication**: if dividends are at most $I(1)$, then without a bubble the price and the price-dividend ratio are **not explosive**
- $I(1)$: integrated of order one, a random walk with stationary increments

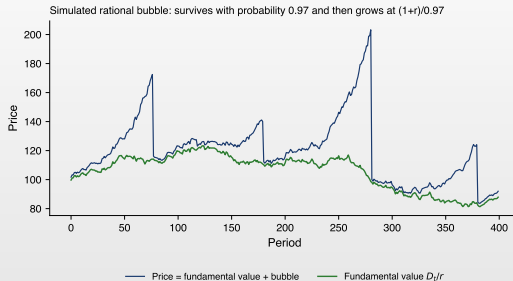
Worked Example: Fundamental Value and Bubble

- Next dividend $D_1 = 2$, required return $r = 6\%$, dividend growth $g = 2\%$: $F = 2/(0.06 - 0.02) = 50$
- Market price $P = 65$: bubble $B = 15$, i.e. 23% of the price
- In 5 years: $F_5 = 50 \cdot 1.02^5 = 55.2$, $\mathbb{E}[B_5] = 15 \cdot 1.06^5 = 20.1$: the bubble share rises to 27%
 - ▶ In 30 years: $F_{30} = 90.6$, $\mathbb{E}[B_{30}] = 86.2$: the bubble is about half of the price
- The same algebra is solved step by step in Seminar 17, Exercise A1

Bubbles That Burst: Blanchard and Watson

- Each period the bubble survives with probability π and collapses with probability $1 - \pi$ [Blanchard & Watson \(1982\)](#)
 - ▶ $B_{t+1} = \frac{1+r}{\pi} B_t + \varepsilon_{t+1}$ with probability π , $B_{t+1} = \varepsilon_{t+1}$ otherwise
- Expected growth is still r ; while it survives the bubble grows faster, at $(1+r)/\pi - 1$, to pay for the crash risk
- Expected lifetime: $1/(1-\pi)$ periods, a geometric waiting time
- Simulation on the next slide: $r = 1\%$ per period, $\pi = 0.97$: growth 4.1% per period while alive, expected life 33.3 periods

A Simulated Rational Bubble



- Fundamental value D_t/r with random-walk dividends; the bubble restarts from a small value after each collapse

Reading the Simulated Bubble

- In 400 periods the bubble grew large and burst 4 times; at its largest the price was 2.06 times the fundamental value
- Each episode: slow start, faster-than-fundamental growth, a sudden fall back to the fundamental value
- Between episodes the price tracks fundamentals: a test on the whole sample sees mostly “normal” data
- This is why the tests of this chapter look at **sub-samples** (windows) rather than at the full sample [Evans \(1991\)](#)

Bubbles Beyond Rational Expectations



Robert J. Shiller, Nobel Prize 2013

Photo: Presidential Office Building, Taiwan (2017);

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- **Feedback and narratives:** rising prices attract new buyers, “new era” stories spread by word of mouth [Shiller \(2015\)](#)
- **Disagreement and short-sale limits:** the price includes the option to resell to a more optimistic buyer [Scheinkman & Xiong \(2003\)](#)
- **Riding the bubble:** informed traders stay in as long as others do, since no one can burst it alone [Abreu & Brunnermeier \(2003\)](#), [Brunnermeier & Nagel \(2004\)](#)
- All three give prices that grow faster than fundamentals for a while: the explosive signature we will test

Fama: Are Bubbles Even Identifiable?

- Fama's objection: a "bubble" is only a bubble if its end is predictable; otherwise it may be a rational reaction to news [Fama \(2014\)](#)
- New technologies raise the uncertainty about future profits and can justify high, volatile prices [Pástor & Veronesi \(2009\)](#)
- Two questions we can answer with data
 - ▶ Do sharp run-ups predict crashes, or low returns? (next section)
 - ▶ Is the price path explosive, and since when? (explosiveness tests)

Recap: What Is a Bubble?

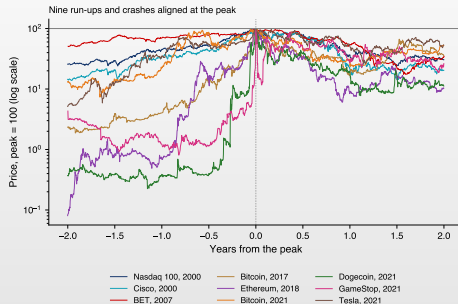
- Bubble = price minus fundamental value, the present value of expected cash flows
- A rational bubble must grow at the required return: its path is explosive, and it is positive
- Bursting bubbles grow even faster while alive; they look normal between episodes
- Behavioural bubbles (feedback, disagreement, riding) share the explosive signature

Nine Episodes: Run-Up and Crash

Episode	Peak	2-year run-up	Max. fall	Trough	Back at peak
Nasdaq 100, 2000	27 Mar 2000	+288%	−83%	Oct 2002	Nov 2015
Cisco, 2000	27 Mar 2000	+592%	−89%	Oct 2002	Aug 2021
BET, 2007	24 Jul 2007	+97%	−83%	Feb 2009	Mar 2021
Bitcoin, 2017	16 Dec 2017	+4186%	−83%	Dec 2018	Nov 2020
Ethereum, 2018	13 Jan 2018	+124004%	−94%	Dec 2018	Feb 2021
Bitcoin, 2021	8 Nov 2021	+667%	−77%	Nov 2022	Mar 2024
Dogecoin, 2021	7 May 2021	+26897%	−92%	Jun 2022	not yet
GameStop, 2021	27 Jan 2021	+2218%	−88%	Apr 2024	not yet
Tesla, 2021	4 Nov 2021	+1837%	−74%	Jan 2023	Dec 2024

- Peak: highest daily close in the episode; run-up: price change over the two years before the peak; maximum fall: to the lowest close within four years
- BET 2007: the only episode without a spectacular run-up in the last two years, yet with one of the deepest falls

Run-Ups and Crashes Aligned at the Peak



□ Daily prices two years before and after each peak, peak = 100, log scale

Reading the Aligned Episodes

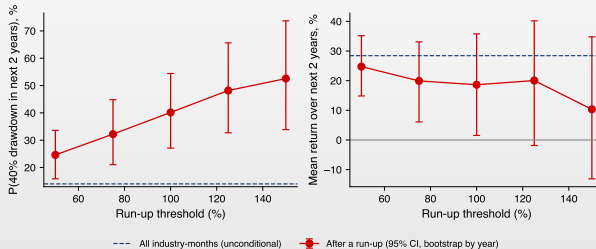
- ▣ Two-year run-ups differ enormously, from +97% (BET) to +124004% (Ethereum); the falls are much more alike, 74–94%
- ▣ The shape is similar: an accelerating rise (convex on the log scale), a sharp top, a deep fall
- ▣ Acceleration on a log scale means the growth rate itself rises: this is the **explosive** behaviour the tests look for
- ▣ Selection bias: we picked these episodes because they crashed; many run-ups did not

Bubbles for Fama: The Question

- Greenwood, Shleifer and You look at **all** US industries since 1926, not at famous episodes only [Greenwood, Shleifer & You \(2019\)](#)
- **Run-up**: value-weighted industry return over the past 2 years above a threshold (e.g. 100%), both raw and net of the market
 - ▶ New episode only if the last one in the same industry started more than 24 months earlier
- **Crash**: a drawdown of at least 40% from a previous high within the next 2 years
- Our replication: 49 industries of the [Kenneth French Data Library](#), 54 805 industry-months, 1926–2026

What Do Sharp Run-Ups Predict?

Bubbles for Fama, 49 US industries 1926-2026: two-year run-ups above the threshold, raw and net of the market



95% CI: bootstrap over calendar years, because run-ups cluster in time (1928–1936, 1999–2001)

MFM_ch17_bubbles_for_fama

Reading Bubbles for Fama

- ▣ After a run-up above 100% (127 episodes in 42 industries), a 40% crash followed in 40% of cases
 - ▶ 95% CI [27%, 54%], against 14% for a random industry-month
 - ▶ Above 150%: 53% ($n = 59$)
- ▣ But the mean return over the next 2 years stays positive: 19% after a 100% run-up, CI [2%, 36%]
 - ▶ Unconditional mean: 28%; the difference is not significant
- ▣ Run-ups raise the **risk** of a crash, not the predictable loss: some keep rising, some collapse
- ▣ This matches Greenwood, Shleifer and You: price run-ups alone do not give a profitable timing signal
Greenwood, Shleifer & You (2019)

The Dot-Com Bubble



The Pets.com sock puppet, mascot of an online pet-supply retailer that listed and closed in 2000

Photo: Atomic Taco (2013); CC BY-SA 2.0; Wikimedia

[Commons](#)

- Nasdaq 100: +288% in the two years to its closing high of 27 Mar 2000, then −83% to Oct 2002
- Cisco: +592% in two years, then −89%; back at the peak only in Aug 2021
- Hedge funds rode the bubble and cut positions before the fall [Brunnermeier & Nagel \(2004\)](#)
- Yet the internet did change the economy: a technology revolution can create both value and a bubble [Pástor & Veronesi \(2009\)](#)

Recap: Episodes

- ▣ Famous bubbles share a shape: accelerating rise, sharp top, deep fall
- ▣ Across all US industries, run-ups above 100% are followed by a 40% crash about 40% of the time vs 14% unconditionally
- ▣ Average returns after run-ups stay positive: run-ups signal risk, not a sure loss
- ▣ Next: formal tests of explosive behaviour on a single series

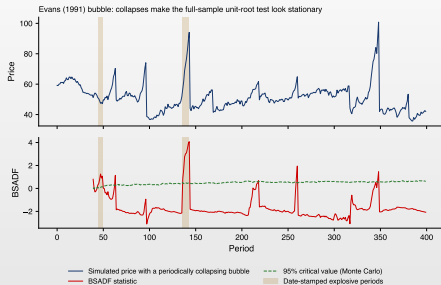
From Unit Roots to Explosive Roots

- Regression on the log price y_t : $\Delta y_t = a + b y_{t-1} + \varepsilon_t$, i.e. $y_t = a + (1 + b) y_{t-1} + \varepsilon_t$
 - ▶ $b < 0$: mean reversion; $b = 0$: random walk (unit root); $b > 0$: **explosive**, deviations grow geometrically
- ADF (augmented Dickey–Fuller) statistic: $ADF = \hat{b}/SE(\hat{b})$; lagged Δy terms may be added
- Usual test: $H_1 : b < 0$ (left tail). Bubble test: $H_0 : b = 0$ vs $H_1 : b > 0$ (**right tail**) Phillips, Wu & Yu (2011)
- In this chapter: no lagged terms, a constant, log prices or log price-dividend ratios

The Problem: Collapsing Bubbles Hide

- ▣ Evans: a bubble that collapses periodically makes the **full-sample** series look stationary or like a random walk
Evans (1991)
- ▣ One regression averages explosive phases with crashes and calm periods: low power
- ▣ Solution: run the ADF regression on many **windows** and take the largest statistic
 - ▶ SADF (Phillips, Wu and Yu): windows start at the first observation and grow Phillips, Wu & Yu (2011)
 - ▶ GSADF (Phillips, Shi and Yu): windows may start and end anywhere Phillips, Shi & Yu (2015a)

Evans Bubble: Full Sample vs Windows

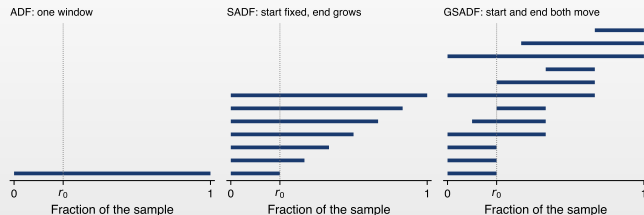


- 400 simulated periods; BSADF (defined below) against its 95% critical value from 2000 random-walk paths

Reading the Evans Simulation

- ▣ Full-sample right-tailed ADF: -5.16 vs 95% critical value -0.09 : no rejection, the bubble is invisible
- ▣ SADF 1.26 (critical value 1.46) and GSADF 4.08 (critical value 2.21)
- ▣ BSADF crosses its critical value in most bursts, but only 2 bursts last long enough to be date-stamped
- ▣ Short explosive bursts are filtered out by the minimum-duration rule: the price paid for fewer false alarms

Three Ways to Choose Windows



- r_0 : the smallest window, as a fraction of the sample; each bar is one ADF regression

SADF, GSADF and BSADF: Definitions

- $ADF_{r_1}^{r_2}$: the ADF statistic on the window from fraction r_1 to fraction r_2 of the sample
- $SADF(r_0) = \sup_{r_2 \in [r_0, 1]} ADF_{r_0}^{r_2}$ Phillips, Wu & Yu (2011)
- $GSADF(r_0) = \sup_{r_2 \in [r_0, 1]} \sup_{r_1 \in [0, r_2 - r_0]} ADF_{r_1}^{r_2}$ Phillips, Shi & Yu (2015a)
- Backward SADF at the end point r_2 : $BSADF_{r_2} = \sup_{r_1 \in [0, r_2 - r_0]} ADF_{r_1}^{r_2}$; $GSADF = \sup_{r_2} BSADF_{r_2}$
 - $BSADF_{r_2}$ uses only data up to r_2 : it can be computed **in real time**
- Minimum window rule: $r_0 = 0.01 + 1.8/\sqrt{T}$, e.g. $T = 440$ months: $r_0 = 0.096$, 42 months Phillips, Shi & Yu (2015a)

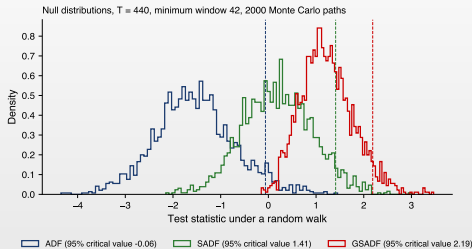
Date-Stamping: When Did It Start and End?

- ▣ **Origination:** the first date at which BSADF exceeds its critical value Phillips, Shi & Yu (2015a)
 - ▶ The critical value sequence cv_{r_2} : 95% quantile of $BSADF_{r_2}$ under the null, date by date
- ▣ **Termination:** the first later date at which BSADF falls back below its critical value
- ▣ **Minimum duration:** keep episodes lasting at least $\log T$ periods (7 months for $T = 440$), to avoid flagging single jumps
- ▣ PWY date-stamping uses the recursive $ADF_0^{r_2}$ instead: it may miss a second bubble after a collapse Phillips, Shi & Yu (2015b)

Worked Example: Right-Tailed ADF Step by Step

- Eight monthly log prices: 4.60, 4.62, 4.66, 4.71, 4.79, 4.90, 5.05, 5.25; regression on 7 changes
- Each change is larger than the one before, and changes grow with the level: $\hat{b} > 0$
- $\hat{b} = 0.392$, $SE(\hat{b}) = 0.017$, $ADF = 22.76$
- Simulated 95% critical value of the right-tailed ADF for 7 observations: 0.24: reject the random walk in favour of an explosive root
- All sums are shown in Seminar 17, Exercise A3

Null Distributions and Critical Values



- Random walk with a small drift, $y_t = T^{-1} + y_{t-1} + \varepsilon_t$, 2000 paths of $T = 440$ Phillips, Shi & Yu (2015a)

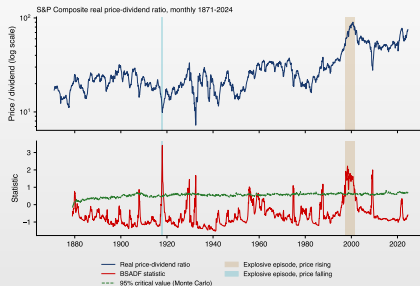
Reading the Null Distributions

- ▣ A single ADF: 95% critical value -0.06 (negative!); SADF: 1.41; GSADF: 2.19
- ▣ Taking a supremum over more windows shifts the distribution to the right: more windows, higher bar
- ▣ The GSADF bar is the price of searching over about 80 000 windows: it controls the chance of a false alarm somewhere in the sample
- ▣ Critical values depend on T , r_0 and the lag order: simulate them for each application

Changing Volatility: Wild Bootstrap

- Monte Carlo critical values assume shocks with constant variance; real prices have volatility clusters and shifts
- A burst of volatility can mimic explosiveness and inflate the rejection rate [Harvey et al. \(2016\)](#)
- **Wild bootstrap:** multiply the observed, demeaned changes Δy_t by random signs $w_t = \pm 1$, rebuild the path, recompute BSADF [Phillips & Shi \(2020\)](#)
 - ▶ The bootstrap null keeps the volatility pattern of the data but has no explosive root
- We use it for Bitcoin, where annualised volatility moved between about 30% and over 100%

S&P 500 Price-Dividend Ratio, 1871–2024

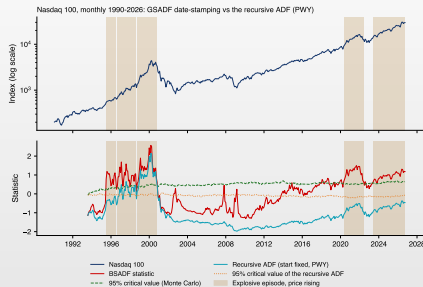


- Real price over real dividend, monthly, 1841 observations; minimum window 95 months; 1000 Monte Carlo paths

Reading the Price-Dividend Ratio

- $GSADF = 3.41$ vs 95% critical value 2.39 (99%: 2.90): explosive episodes exist; the full-sample ADF (-1.37) sees none
- Date-stamped episodes: Aug 1917–May 1918; May 1997–Jul 2001
- The dot-com episode is the dominant one: the ratio peaked at 90.2 in Aug 2000, against a long-run mean of 29.1
 - ▶ 1917–1918 is an accelerating **fall** of the ratio: real prices fell faster than real dividends; explosive, but not a bubble
- 1879–1880, 1929, 1955, 1987 and 2008–2009: BSADF crossed the critical value, but for fewer than 8 months in a row: filtered out by the minimum duration
- Explosiveness in a ratio can come from the denominator: always check both price and dividend

Nasdaq 100: GSADF vs PWY Date-Stamping



- Monthly log closes 1990–2026, $T = 440$, minimum window 42 months; 2 000 Monte Carlo paths

Reading the Nasdaq 100

- $GSADF = 2.57$ (95% critical value 2.19); $SADF = 2.14$ (critical value 1.41)
- BSADF episodes: Jun 1995–Jun 1996; Aug 1996–Jul 1998; Sep 1998–Oct 2000; Apr 2020–May 2022; May 2023–ongoing
 - ▶ PWY (fixed start) episodes: Apr 1997–Nov 1997; Jan 1998–Jul 1998; Sep 1998–Oct 2000
- The recursive ADF starts in 1990 and is dragged down by the long flat period after 2000: it misses the post-2020 episodes that BSADF finds
- The dot-com signal started in Jun 1995, 57 months before the peak of Mar 2000

Bitcoin: Explosive Episodes

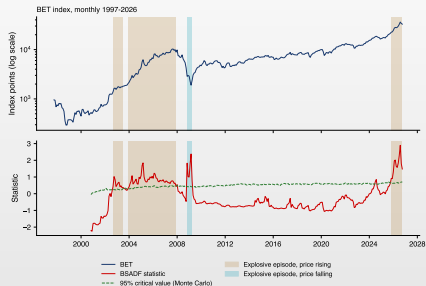


- Weekly log closes 2014–2026, $T = 626$, minimum window 51 weeks; Monte Carlo critical values

Reading Bitcoin

- GSADF = 3.85: Monte Carlo 95% critical value 2.27, wild bootstrap 2.82: rejected either way
- Monte Carlo episodes: Apr 2017–Jun 2018; Dec 2020–May 2021
 - ▶ Wild bootstrap episodes: May 2017–Jan 2018; Dec 2020–Feb 2021
- 2017: signal from Apr 2017, 8 months before the peak of 16 Dec 2017; it ended 6 months after the peak
- The 2024–2025 rise to new highs is **not** flagged: steady growth is not the same as explosive acceleration

BET: The 2007 Boom and 2025

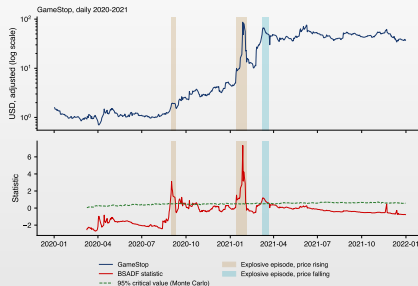


- Monthly log closes of the official BET, 1997–2026, $T = 348$, minimum window 37 months

Reading the BET

- $GSADF = 2.90$ vs 95% critical value 2.21; episodes: Aug 2002–Jun 2003; Nov 2003–Oct 2007; Oct 2008–Mar 2009; Oct 2025–ongoing
- 2003–2007: the run-up before and around Romania's accession to the European Union (January 2007); BET then fell 83% from its peak of 24 Jul 2007 [Pele & Mazurencu-Marinescu \(2012\)](#)
- Herding and LPPL patterns on the Bucharest Stock Exchange in that period: [Pele, Mazurencu & Nijkamp \(2013\)](#)
- The current episode (from Oct 2025) is still open: explosive does not mean it will crash; it means the rise is accelerating

GameStop, January 2021



□ Daily log closes 2020–2021, $T = 504$, minimum window 45 days

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Reading GameStop



A GameStop store, 3 February 2021, a week after the peak

Photo: Will Buckner (2021); CC BY 2.0; Wikimedia Commons

- $GSADF = 7.34$, far above its critical value 2.21
- Episodes: 31 Aug 2020–9 Sep 2020; 13 Jan 2021–3 Feb 2021; 8 Mar 2021–22 Mar 2021
- The January signal came on 13 Jan 2021, 14 days before the closing high of 27 Jan 2021
- A short squeeze, not a slow bubble: the explosive phase lasted weeks, and daily data are needed to see it

Real Time vs Hindsight

Episode	First signal Peak	Lead (months)	Rise after signal	Fall after peak
Nasdaq 100, monthly	Jun 1995 / Mar 2000	57	+774%	−76%
BET, monthly	Nov 2003 / Jul 2007	44	+429%	−83%
Bitcoin 2017, weekly	Apr 2017 / Dec 2017	8	+1381%	−83%
Bitcoin 2021, weekly	Dec 2020 / Apr 2021	4	+157%	−75%
GameStop, daily	13 Jan 2021 / 27 Jan 2021	0.5	+1007%	−88%

- First signal: start of the chain of BSADF episodes that leads to the peak; rise: from the close on the signal date to the peak; fall: to the lowest close within 2 years (GameStop: 1 year)
- In real time the tests **warn early**, but the lead ranges from 2 weeks to almost 5 years, and selling at the first signal would have missed most of the rise: they do not time the crash [Phillips, Shi & Yu \(2015b\)](#)

Recap: Explosiveness Tests

- Right-tailed ADF on log prices or ratios; bubbles show up as $b > 0$ in some windows
- GSADF tests for at least one episode; BSADF date-stamps them in real time, with a minimum duration
- Critical values by simulation; wild bootstrap when volatility changes
- Explosive does not mean irrational: fundamentals can accelerate too; the price-dividend ratio controls for dividends

Crashes as Critical Points



Didier Sornette, ETH Zurich

Photo: Didier Sornette (2012); CC BY-SA 3.0
(Germany); Wikimedia Commons

- Johansen, Ledoit and Sornette (JLS): imitation among traders makes the **hazard rate** of a crash rise as a critical time t_c approaches [Johansen, Ledoit & Sornette \(2000\)](#)
- Rational investors stay in only if the price grows fast enough to pay for the rising crash risk
- Result: **faster-than-exponential** growth, decorated by oscillations that speed up near t_c
- t_c is the most probable end of the bubble, not a certain crash date [Sornette \(2017\)](#)

The LPPLS Equation

- ▣ $\ln p(t) = A + B(t_c - t)^m + C(t_c - t)^m \cos(\omega \ln(t_c - t) - \phi), \quad t < t_c$
- ▣ A : log price at t_c ; $B < 0$: the super-exponential trend for a rising bubble; $0 < m < 1$: power-law exponent
 - ▶ The growth rate $-Bm(t_c - t)^{m-1}$ rises without bound as $t \rightarrow t_c$
- ▣ C, ω, ϕ : amplitude, angular log-frequency and phase of the log-periodic oscillations
- ▣ Seven parameters: A, B, C, ϕ and the nonlinear t_c, m, ω
- ▣ LPPL: the log-periodic power law pattern; LPPLS: the model with the finite-time singularity at t_c

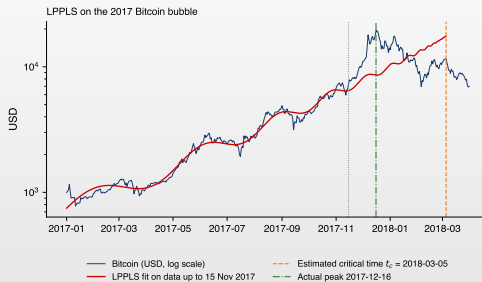
Calibration: Filimonov and Sornette

- Rewrite the cosine: $C \cos(\omega \ln(t_c - t) - \phi) = C_1 \cos(\omega \ln(t_c - t)) + C_2 \sin(\omega \ln(t_c - t))$ Filimonov & Sornette (2013)
 - ▶ A, B, C_1, C_2 are now **linear**: for given (t_c, m, ω) they come from OLS
- Search only over (t_c, m, ω) : a grid, then a Nelder–Mead refinement of the sum of squared residuals
- Our search ranges: t_c up to half the window length after its end, $m \in [0.01, 0.99]$, $\omega \in [2, 25]$
- Many local minima remain: different windows and starting points give different t_c Geraskin & Fantazzini (2013)

Qualified Fits: The Filter

- ▣ A fit is kept only if it describes a real accelerating bubble, in the spirit of Sornette and co-authors [Demirer et al. \(2019\)](#)
- ▣ Our filter: $B < 0$; $0.01 \leq m \leq 0.99$; $2 \leq \omega \leq 15$; t_c at most 5% of the window before and 10% after its end
 - ▶ Number of oscillations $O = \frac{\omega}{2\pi} \ln \frac{t_c - t_1}{t_c - t_2} \geq 2.5$, so the oscillations are not a single bump
 - ▶ Damping $D = \frac{m|B|}{\omega|C|} \geq 0.5$, so the hazard rate stays positive
- ▣ $[t_1, t_2]$: the estimation window; the filter is applied to each window separately

LPPLS on Bitcoin, 2017

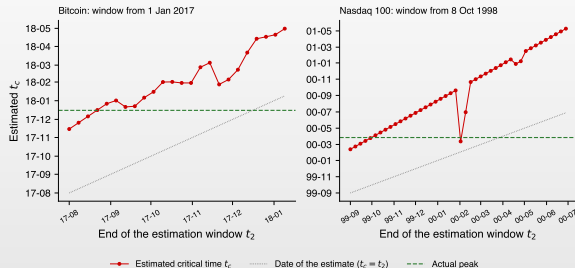


- Fit on daily log prices from 1 January to 15 November 2017; the path after 15 November is out of sample

Reading the Bitcoin Fit

- ▣ The fit follows the 2017 rise and its oscillations closely: $m = 0.91$, $\omega = 16.88$, $B = -2.79$
- ▣ Estimated t_c : 5 Mar 2018; actual peak: 16 Dec 2017, i.e. 79 days earlier
- ▣ ω is above 15: this fit does **not** pass the filter, even though it looks convincing
- ▣ A good in-sample fit is not evidence of a forecasting ability: we must look at many fits made in real time

How Stable Is the Critical Time?



- Each point: one fit on a window with fixed start, ending on the date on the horizontal axis (every 7 days)

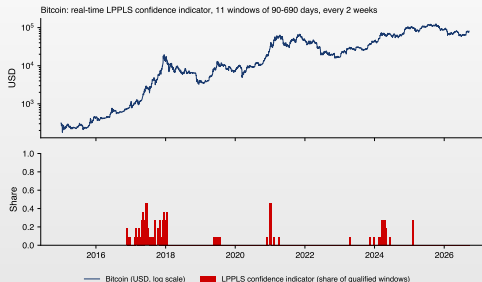
Reading the Critical-Time Paths

- Bitcoin: before the peak (20 fits) t_c ranged from Nov 2017 to Mar 2018; mean absolute error 37 days
 - ▶ Within 30 days of the true peak: 45% of the fits
- Nasdaq 100: before the peak (30 fits) t_c ranged from Feb 2000 to Dec 2000; mean absolute error 106 days
 - ▶ t_c keeps moving ahead as data arrive, and locks onto the peak only **after** the peak
- Qualified fits before the peak: 0% (Bitcoin) and 0% (Nasdaq 100): the filter rejects all these single-window fits
- Demos and Sornette: the birth of a bubble is easier to diagnose than its burst [Demos & Sornette \(2017\)](#)

The LPPLS Confidence Indicator

- ▣ At each date t_2 , fit many windows $[t_2 - L, t_2]$ and report the **share of qualified fits** [Demirer et al. \(2019\)](#)
 - ▶ Here: 11 window lengths L from 90 to 690 trading days, every 2 weeks; data up to t_2 only
- ▣ A high share means the bubble signature is robust to the choice of window
- ▣ Honest evaluation: is a signal followed by a fall of at least 20% within 90 days more often than no signal?
 - ▶ Fall measured from the price on the signal date

Bitcoin: Real-Time Confidence Indicator



- Indicator computed every 2 weeks, 2016–2026; bar height: share of the 11 windows with a qualified fit

Reading the Confidence Indicator

- Bitcoin: the indicator was positive on 81% of the 2017 dates, 12% in 2021 and 11% in other years; maximum 0.45 in Jun 2017
- After a signal, a 20% fall within 90 days followed in 37% of cases, vs 38% without a signal (257 dates)
 - ▶ Nasdaq 100, 1997–2026: 3% after a signal vs 8% without (534 dates)
- In our data the indicator identifies accelerating rises, but it does **not** predict large falls better than chance
- Results depend on our filter and windows; published indicators use more windows and a different filter [Demirer et al. \(2019\)](#)

LPPL in the Cryptocurrency and Romanian Markets

- ▣ Bucharest Stock Exchange: herding and log-periodic patterns before the 2007–2008 crash [Pele, Mazurencu & Nijkamp \(2013\)](#); crash modelling for the BVB [Pele & Mazurencu-Marinescu \(2012\)](#)
- ▣ Cryptocurrencies: a fundamental value from Metcalfe's law (network size) combined with LPPL bubbles [Pele & Mazurencu-Marinescu-Pele \(2019\)](#), [Conda et al. \(2026\)](#)
- ▣ Dogecoin 2021: LPPL together with sentiment from social media [Pele \(2025\)](#)
- ▣ Common lesson: LPPL describes the acceleration well; the crash date remains the weak point

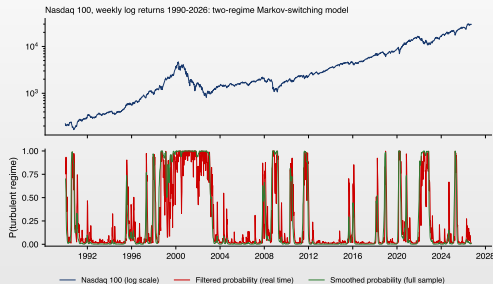
Recap: LPPLS

- Faster-than-exponential growth with accelerating oscillations towards a critical time t_c
- Filimonov–Sornette: four linear parameters by OLS, three nonlinear by search
- Single fits look convincing but t_c moves with the window; many fits fail the filter
- Always evaluate in real time: in our data the confidence indicator had no edge for predicting 20% falls

The Markov-Switching Model

- Weekly log return $r_t = \mu_{s_t} + \sigma_{s_t}\varepsilon_t$, $\varepsilon_t \sim N(0, 1)$, regime $s_t \in \{1, 2\}$ unobserved [Hamilton \(1989\)](#)
 - ▶ Each regime has its own mean μ and volatility σ : a calm and a turbulent regime
- Regimes follow a Markov chain: $p_{ij} = P(s_t = j \mid s_{t-1} = i)$; expected duration of regime i : $1/(1 - p_{ii})$
- Estimation by maximum likelihood with the Hamilton filter, which updates $P(s_t = j \mid r_1, \dots, r_t)$ week by week
- **Filtered** probability: uses data up to t (real time); **smoothed**: uses the whole sample (hindsight)
 - ▶ A worked filter step is in Seminar 17, Exercise A5

Nasdaq 100: Two Regimes



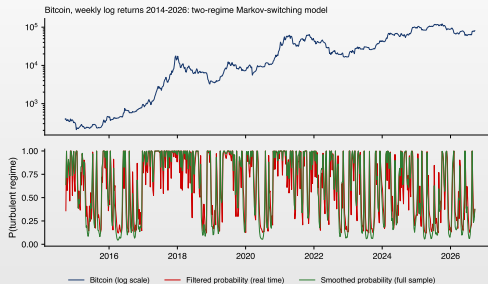
- Weekly log returns 1990–2026, 1 915 weeks; regime means and variances both switch

Nasdaq 100: Estimated Regimes

Regime	$\hat{\mu}$ (% per week)	SE	$\hat{\sigma}$ (%)	$\hat{p}_{ij}$	Duration (weeks)
Calm	0.43	0.06	2.30	0.987	75.4
Turbulent	-0.32	0.27	5.40	0.957	23.4

- Calm regime: positive drift, low volatility, 76% of the weeks; turbulent: volatility more than twice as high, negative (insignificant) mean
- The turbulent spell around the 2000 peak began in Jun 1998 (smoothed probability above 0.5); in real time the filtered probability was already 0.99 at the end of March 2000
- The last phase of the dot-com bubble was already turbulent: prices kept rising while volatility jumped

Bitcoin: Two Regimes



- Weekly log returns 2014–2026, 626 weeks

Reading the Bitcoin Regimes

- Calm: $\hat{\mu} = 0.26\%$ per week, $\hat{\sigma} = 2.93\%$, duration 3.8 weeks
 - ▶ Turbulent: $\hat{\mu} = 1.21\%$ (SE 0.58), $\hat{\sigma} = 11.22\%$, duration 6.3 weeks, 62% of the weeks
- Unlike equities, Bitcoin's turbulent regime has the **higher** mean: booms are as turbulent as busts
- Regimes are short and switch often: they describe volatility states, not bubble phases
- A three-regime model is compared in Seminar 17, Exercise B8

What Regimes Can and Cannot Tell

- ▣ They date turbulent periods in real time (filtered probability) and give expected durations
- ▣ They do not use the level of prices: a slow, calm bubble stays in the calm regime
- ▣ Combine them with explosiveness tests: BSADF says “the price is accelerating”, the regime says “risk has changed”
- ▣ Regime models feed risk management: VaR and ES by regime (Chapters 7–8)

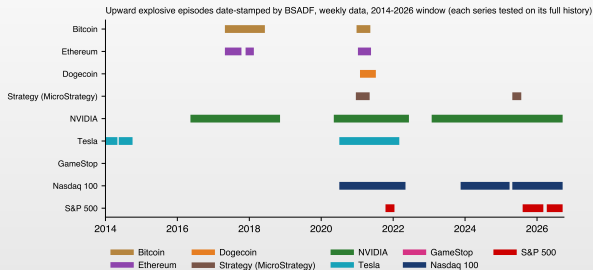
Recap: Regimes

- ▣ Two regimes with different mean and volatility, linked by a Markov chain [Hamilton \(1989\)](#)
- ▣ Nasdaq 100: long calm regime (about 75.4 weeks), shorter turbulent one; the 2000 top happened in the turbulent regime
- ▣ Bitcoin: short regimes, the volatile one with the higher mean return
- ▣ Only filtered probabilities are available in real time

Are Crypto-Assets Bubbles?

- No dividends: the fundamental value of Bitcoin is hard to define; one estimate puts it at zero [Cheah & Fry \(2015\)](#)
- Alternative: value from network adoption (Metcalf's law), with LPPL bubbles on top [Pele & Mazurencu-Marinescu-Pele \(2019\)](#)
- Crypto-assets increasingly behave like an alternative asset class, not a pure bubble [Pele et al. \(2023\)](#)
- Without a fundamental value we can still test the price: is it explosive, and when?

Explosive Episodes, 2014–2026



□ Weekly log closes; each series tested on its own full history with its own Monte Carlo critical values; rising episodes only

 MFM_ch17_crypto_ai

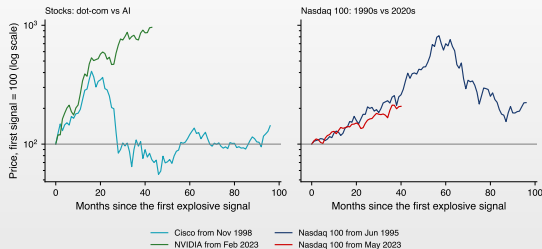
Reading the Timeline

- Crypto: Bitcoin Apr 2017–Jun 2018; Dec 2020–May 2021; Ethereum Apr 2017–Oct 2017; Nov 2017–Feb 2018; Jan 2021–May 2021; Dogecoin Jan 2021–Jul 2021
 - ▶ Crypto booms cluster in time: 2017 and late 2020–2021, across all three coins
- Strategy, a company that holds Bitcoin: Dec 2020–May 2021; Apr 2025–Jul 2025; Tesla: Nov 2013–May 2014; May 2014–Oct 2014; Jul 2020–Mar 2022
- NVIDIA: May 2016–Nov 2018; May 2020–Jun 2022; Jan 2023–ongoing
- GameStop, weekly: the January 2021 squeeze lasted only 3 weeks, shorter than the minimum duration: the weekly test misses what the daily test sees

Explosive Up or Explosive Down?

- ▣ $b > 0$ means that the deviations grow; it does not say in which direction
- ▣ An accelerating **fall** also gives $b > 0$: large negative changes at low prices
- ▣ Examples in our data: BET, October 2008–March 2009; Nasdaq 100 after the 2000 peak
- ▣ Always report the price change over each episode; the charts shade rising and falling episodes differently

Dot-Com vs AI



- Monthly closes, rebased to 100 at the first explosive signal of each episode (BSADF, monthly data)

Reading Dot-Com vs AI

- Cisco: first signal Nov 1998, peak after 16 months at 4.1 times the signal price
 - ▶ NVIDIA: first signal Feb 2023; 43 months later at 9.6 times the signal price, and the episode is still open
- Nasdaq 100 from Jun 1995: peak after 57 months at 8.2 times the signal level
 - ▶ Nasdaq 100 from May 2023: 2.1 times the signal level after 40 months: so far a slower rise than in the 1990s
- NVIDIA's price path is at least as explosive as Cisco's; the index is much less so
- Open question for Seminar 17, Part C: is this explosiveness in prices or also in earnings?

Is There an AI Bubble? What the Tests Can Say

- Monthly BSADF flags NVIDIA (Mar 2016–Nov 2018; Feb 2020–May 2022; Feb 2023–ongoing) and the Nasdaq 100 (current episode from May 2023)
- The tests cannot separate a bubble from explosive fundamentals: fast-growing expected profits of a new technology can also make prices explosive [Pástor & Veronesi \(2009\)](#)
- History: Cisco's dividend-adjusted share price returned to its March 2000 level only in Aug 2021
- What we can say: prices are in an explosive phase, crash risk is elevated (Bubbles for Fama), the timing is unknown

Recap: Crypto and AI

- ▣ Crypto booms of 2017 and 2021 are clearly explosive; the 2024–2025 rise is not
- ▣ Frequency matters: short squeezes need daily data, long booms monthly data
- ▣ Explosive can mean up or down: always check the direction
- ▣ AI stocks are in an explosive phase; the tests cannot say whether fundamentals justify it

An Open Question

Can bubble alarms be raised in real time with a controlled false-alarm rate, and do they carry crash information?

- BSADF date-stamping runs one right-tailed test per date: the size is controlled date by date, not over the whole monitoring period
 - ▶ Live research: monitoring rules for bubbles *and* crashes with controlled false alarms Whitehouse, Harvey & Leybourne (2023); real-time “econometric thermometers” for housing Shi & Phillips (2023)
- Formal version, for an alarm A_t and a crash $C_{t,h}$ (a fall of at least 20% within $h = 90$ days):
 $H_0: P(C_{t,h} | A_t = 1) = P(C_{t,h} | A_t = 0)$, H_1 : larger after an alarm
 - ▶ Falsified if the hit rate after alarms does not exceed the rate without alarms, under inference that respects overlapping windows
- Identification: explosive prices need not be a bubble
 - ▶ Explosive expected cash flows of a new technology leave the same signature Pástor & Veronesi (2009); run-ups raise crash risk without predictable average losses Greenwood, Shleifer & You (2019)

The Discovery Loop, Step by Step (1/2)

- **Literature:** map what is known, then verify every reference
 - ▶ “Which papers control the false-alarm rate of sequential bubble monitoring, and what size and power do they report? Give DOI and simulation design.”
 - ▶ Resolve each DOI on Crossref and find the simulation table the answer quotes; search with Semantic Scholar, Elicit or Google Scholar
- **Hypotheses:** competing explanations, each with its own observable
 - ▶ “For the current NVIDIA episode, list competing explanations (rational bubble, explosive expected cash flows, falling discount rate) and the observable each one predicts.”
 - ▶ E.g. a bubble predicts an explosive price-to-earnings ratio; explosive cash flows predict explosive earnings with a stable ratio
- **Data and code:** on the course data, replicating a known number first
 - ▶ “Implement GSADF with $r_0 = 0.01 + 1.8/\sqrt{T}$ on weekly Bitcoin closes from data/market and simulate its 95% critical value.”
 - ▶ Replicate first: GSADF = 3.85 and critical value 2.27 on 626 weeks, as in this chapter

The Discovery Loop, Step by Step (2/2)

- **Robustness**, fixed before the alarms are evaluated
 - ▶ “Pre-register the evaluation of the alarm: horizons 30/60/90/180 days, crash thresholds 10/20/30%, inference robust to overlapping windows, Holm correction over the 12 cells.”
 - ▶ Twelve cells tried and one reported is data snooping; twelve cells pre-registered and corrected is a test
- **Critique**: ask the assistant to attack the result
 - ▶ “Act as a referee: why could an alarm-crash association be an artefact of overlapping windows, look-ahead or the choice of assets?”
 - ▶ Expect: dependent observations inflate significance; full-sample critical values; famous episodes only, the selection that Greenwood, Shleifer & You (2019) avoid
- **Report**: hit rates with valid p -values, all 12 cells, the alarms that failed
 - ▶ The prompts that shaped the design go into AI_USE.md

What the Human Must Check

- **Full-sample information in date-stamping:** critical values, lag order or detrending fitted on the whole sample
 - ▶ Test: cut the sample at t and recompute; the flag at t must not change
- **Wrong tail:** left-tail ADF critical values (about -2.86) used against an explosive alternative
 - ▶ Test: on simulated random walks the right-tailed test must reject in about 5% of the paths
- **Changing volatility:** Monte Carlo critical values under constant variance distort the size
 - ▶ Test: wild bootstrap critical values; for Bitcoin 2.82 instead of 2.27
- **Multiplicity:** many dates, assets and windows; some alarm is almost certain
 - ▶ Test: run the monitoring rule on simulated random walks and count the paths with at least one alarm
- **Overlapping outcomes:** 90-day windows overlap, so Fisher's exact test or a χ^2 test overstates significance
 - ▶ Test: block bootstrap or circular rotations of the alarm series, which keep the dependence of both series

Mini-Case: Does the LPPLS Alarm Predict Crashes?

- A typical assistant proposal: “a positive LPPLS confidence indicator announces a crash; compare hit rates with Fisher’s exact test”; alarms come in runs and outcomes overlap, so we test by circular rotations
- Bitcoin (257 dates, every 2 weeks): a fall of 20% within 90 days after 37% of alarms vs 38% without
 - ▶ Difference -1.4 percentage points; one-sided $p = 0.61$ from 206 rotations of the alarm series
 - ▶ Nasdaq 100: 3% vs 8%, difference -4.9 points, $p = 0.89$
- Pre-registered grid of 4 horizons $\times$ 3 thresholds: smallest raw p is 0.38 (Bitcoin) and 0.46 (Nasdaq 100)
 - ▶ No cell rejects H_0 at 5%, before or after the Holm correction
- Interpretation: the indicator dates accelerating rises but carries no information about the timing of large falls in our data

Your Turn: a Project Seed

- **Question:** can a monitoring rule for Bitcoin, the Nasdaq 100 and NVIDIA keep the probability of any false alarm at 5% over the whole period and still warn before large falls?
- **Replicate first:** GSADF = 3.85 for weekly Bitcoin, then the crash probability after run-ups above 100% in [Greenwood, Shleifer & You \(2019\)](#) (ours: 40%)
 - ▶ Extend: calibrate the alarm threshold by simulation over the whole monitoring period [Whitehouse, Harvey & Leybourne \(2023\)](#); evaluate the alarms with rotations or a block bootstrap
- **Deliverables:** a pre-registered grid committed before the evaluation, a notebook that regenerates every number, false-alarm rates and hit rates with valid p -values
- **AI log:** prompts that shaped the design in AI_USE.md; each wrong claim (left-tail critical values, a full-sample flag) and the test that caught it in AI_ERRORS.md ([templates](#))

Summary

- ▣ A bubble is price minus fundamental value; rational bubbles must be explosive, and bursting bubbles hide from full-sample tests
- ▣ GSADF tests for explosive episodes, BSADF date-stamps them in real time; simulate critical values, use the wild bootstrap if volatility changes
- ▣ Across our episodes the tests warned months before the peak, but did not time the crash
- ▣ LPPLS describes the acceleration; its critical time moves with the window, and our confidence indicator did not predict large falls
- ▣ Sharp run-ups raise crash risk without predictable average losses: a risk-management signal, not a trading rule

Which Tool When?

Question	Tool
Has there been any explosive episode?	GSADF with simulated critical values
When did it start and end?	BSADF date-stamping, minimum duration $\log T$
Volatility changes a lot (crypto, crises)	Wild bootstrap critical values
Is the rise accelerating super-exponentially?	LPPLS confidence indicator over many windows
Has the risk state changed?	Markov-switching filtered probabilities
How likely is a crash after a run-up?	Run-up conditioning across many assets (Bubbles for Fama)

Coming Next: Systemic Risk (Chapter 18)

- When a bubble bursts, losses spread through banks and markets: contagion and systemic risk
- CoVaR and SRISK measure how much one institution adds to the risk of the system
- Further reading: [Franke, Härdle & Hafner \(2019\)](#), [Kindleberger & Aliber \(2005\)](#), [Shiller \(2015\)](#), [Sornette \(2017\)](#)

Further Study on Quantinar

Advanced courses on quantinar.com

- ☐  Cryptocurrency: Speculative Asset and Medium of Exchange
- ☐  Are Cryptos becoming alternative Assets?
- ☐  Statistics of Financial Markets

Code and data

- ☐ Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_17

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